



OPEN Cost-effective and scalable urban air quality monitoring using image-based deep learning

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Air pollution is a persisting problem for the environment, mainly in urban areas, given heavy industrialization and vehicular emissions and urbanization. Exposure to harmful air pollutants, including PM_{2.5}, PM₁₀, and toxic gases, is a major contributor to respiratory and cardiovascular diseases, accounting for millions of deaths globally each year. Answer monitoring is, therefore, the first effective measure in limiting the health effects of air pollution. The idea of this study is to advance an effective, scalable image-based deep learning (DL) system to detect air pollution levels within urban areas. The model integrates CBAM with InceptionV3 to enhance feature extraction, improving accuracy and robustness in detecting varying levels of air pollution from images. The proposed work uses Air Pollution Image Dataset from Kaggle, containing urban pollution images from India and Nepal. Preprocessing includes Gaussian Filtering to reduce noise, Histogram Equalization (HE) to enhance contrast, Thresholding to segment image, and resizing to standard dimensions. The model employs InceptionV3 integrated with Convolutional Block Attention Module (CBAM) for enhanced feature extraction. This approach classifies images into three pollution levels: Moderate, Unhealthy for Sensitive Groups (USG), and Unhealthy, enabling efficient air quality monitoring. The CBAM-InceptionV3 model achieved strong performance, with precision, recall, and F1-scores exceeding 0.95 for each class. The proposed CBAM-InceptionV3 model achieves an overall classification accuracy of 95.23%, with precision, recall, and F1-score values consistently above 95%, demonstrating its effectiveness for air pollution level classification.

Keywords Air pollution classification, Deep learning, CBAM, InceptionV3, Urban air quality

Air pollution, especially in large urban areas—the prime environmental issue today—is a main threat to human health^{1,2}. There has been a sharp rise in pollutants in atmosphere due to rapid rates of industrialization, urbanization, and automobile pollution^{3,4}. These pollutants include well particulates in which have been implicated to respiratory and cardiovascular diseases and premature mortality^{5,6}. According to WHO, millions of people die every year because of air pollution and 90% of world populace is showing to air pollutant levels exceeding its guidelines^{7,8}. This hence calls for urgent efforts to develop efficient and scalable air quality management systems capable of detecting the sources of pollution and developing policies for alleviating the effects of pollution especially in most densely populated cities^{9,10}.

Pollution in the air may have both natural and anthropogenic causes¹¹. Human activities such as transport, industries, power generation, deforestation, and agriculture remain the major causes for urban air pollution^{12,13}. Through these activities the air quality is greatly impaired by release of high concentrations of particulate matter and gases¹⁴. Moreover, pollution is worsened by the weather conditions, especially by presence of temperature inversion, especially in crowded urban areas^{15,16}. Natural causes such as wildfire, volcanic eruptions and dust storms also pollute, although they do not have a significant impact compared to those caused by human beings^{17,18}. In cities, the population density, traffic jams, industries, and ineffective environmental policies causes high concentrations of harmful pollutants^{19,20}.

Due to increased demand of air pollution monitoring, different techniques have been devised to measure the level of pollution starting with sensor-based techniques up to image-based techniques^{21,22}. The conventional sensor-based technique is based on physical air quality monitoring stations, which associate the attention of

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pollutants in the air^{23,24}. They are efficient and, in most cases, they prove to be costly, geographically scaled and difficult to implement at the large scale^{25,26}. The machine learning (ML) that are usually used to classify images according to air quality are convolutional neural networks (CNNs), support vector machines (SVM), and decision trees^{27,28}. Although these techniques are more scalable and less expensive, they are also limited by the fact that they do not pay much attention to important image characteristics and that it is hard to distinguish between slight changes in level of pollution²⁹.

The suggested model eliminates them by introducing the CBAM to InceptionV3, which is one of strongest DL models used to classify images. CBAM increases the relevance of the model with respect to the most significant areas in images and extends its capability of detecting the additional features that are subtle clues of particular levels of pollution such as haze, smog or mist. CBAM helps the model to learn automatically about what parts in the image are needed to classify it, effectively resolving the problem of partial feature extraction in other CNN models. InceptionV3 has been used to offer a sophisticated architecture that can recognize complicated details of polluted images without expensive costs and low-performance. The integration of CBAM with InceptionV3 suggests such a framework with ability to be very much accurate as far as the air pollution level prediction is concerned, which is an enhancement over current method. The proposed model contributes to the improvement of classification accuracy, and meanwhile, presents a more scalable and efficient way for monitoring air quality over urban areas. It introduces an image-based machine learning solution for measuring urban air quality via a monitoring system that consists of time-series sensor data in addition to the traditional methods for measurement. The overall model implements CBAM (Convolutional Block Attention Module) and InceptionV3 architecture to create better representations of spatial and channel features and can detect small visual elements such as haze and smog. Major contributions of this paper are:

- The primary goal of the above-presented framework is to apply DL techniques to classify air pollution levels in the city with images. The dataset used is from the Air Pollution Image Dataset presented on Kaggle containing different labeled images of air pollution levels.
- The framework applies various image preprocessing techniques, including Gaussian filtering to remove noise, HE to enhance contrast, thresholding to segment objects of interest, and image resizing to standardize the image dimensions. Additionally, label encoding is used to change categorical pollution levels into numerical values for the DL model.
- The dataset is divided into training, validation, and test sets, ensuring a stratified split to preserve the proportion of each pollution level in all subsets. This approach ensures that each data split represents all classes fairly, preventing class imbalance during training, validation, and testing.
- The framework uses InceptionV3 for feature extraction and classification from images, integrating CBAM to enhance focus on critical regions like haze and smog. This attention mechanism improves the model's ability to exactly classify air pollution levels.
- A structured ablation study is conducted to quantify the individual contributions of preprocessing, augmentation, and CBAM attention in improving classification performance.

The organization of this paper is: “[Literature survey](#)” section reviews existing literature on air pollution classification methods. “[Proposed methodology](#)” section describes the proposed methodology. “[Result and discussion](#)” section discusses the results, evaluating model performance. “[Conclusions and future scope](#)” section concludes the conclusion and future research to improve air pollution classification.

Literature survey

Sarkar et al.³⁰ studied air quality foretelling for PM2.5 pollution in Delhi by comparing different forecasting models on various criteria. This included comparing capabilities of independent ML and DL models to be used for this prediction. The criteria for judging proficiency of models include R2, MAE, and RMSE. Hybrid-LSTM-GRU was best model in terms of efficiency are used for prediction of AQI values from all other models. Das et al.³¹ have analyzed air quality prediction methods for PM10 and SO2 pollutants in Basaksehir district of Istanbul using methods LSTM, RNN, and MLP. The study also looks into training using incomplete data and revised error terms. Overall, LSTM least generated errors together with values close to actual ones for the prediction of PM10 and SO2 compared to both MLP and RNN. So, LSTM were performing a lot better than many other studies in field when predicting air quality.

Maltare & Vahora³² compared various machine-learning algorithms for predicting Ahmedabad, Gujarat, India's AQI: SARIMA, SVM, and LSTM. The data were obtained from the Central Pollution Control Board of India, and preprocessing was carried out in different ways for each model. The SVM with RBF kernel has besides other kernels served as the most accurate predictor method for AQI in Ahmedabad compared to SARIMA and LSTM. Gupta et al.³³ performed various types of research on different methods through which air quality can be predicted using data mining techniques for Indian cities are used for this research. The dataset used in present study comprised cities like New Delhi, Bangalore, Kolkata, and Hyderabad. To counter the problem of imbalance in dataset, an oversampling approach named Synthetic Minority Oversampling Technique (SMOTE) was used. The effect of SMOTE to improve the models' performance is well documented together with detailed metrics and visual comparisons so as to foster similar attempts by future researchers.

Heydari et al.³⁴ have developed a hybrid intelligent model, which could effectively predict the emissions from Combined Cycle Power Plants by means of (LSTM) and Multi-Verse Optimization (MVO). The LSTM model predicts the pollutants. On other hand, the MVO enhances LSTM parameters in order to minimize the prediction error. The proposed model was tested against actual data from Kerman power plant in Iran in reference to meteorological input and lagged levels of pollutants. Drewil & Al-Bahadili³⁵ enumerated a model that encompasses some DL methods and ML methods used for the enhancement of air quality forecasting. The

objective of this model involves detecting the most excellent hyperparameters for LSTMs in terms of predicting PM10s, PM2.5s, CO, and NOx following day. The GA optimizes LSTM parameters which help overcome challenges faced, such as dealing with noise and setting hyperparameters incorrectly. Results show that there was an outperformance of proposed GA-LSTM model over conventional ML and LSTM models, with greater accuracy, speed, and efficiency.

Kumar & Pande³⁶ on air pollution data of 6 years of 23 cities of India, pre-processed data set, which is selecting important features through correlation analysis. The reduction in pollution was pretty visible in this pandemic year 2020. Due to the present situation of data imbalance, combinations of resampling techniques were applied, and five ML models were used. It was found that the accuracy produced by XGBoost was the greatest from all the others, and linearly good was the prediction with the actual data. Samad et al.³⁷ used ML techniques to simulate the concentrations of five models; ridge regression, support vector regression, random forests, extra trees regression, and extreme gradient boosting. The study concluded that predictions were considerably influenced by nearby pollutant data and that the methodology was extendable to other locations, as was demonstrated with Karlsruhe-Nordwest data. Ahmad and Kumar³⁸ proposed a hybrid WaveNet-based deep learning framework enhanced with XGBoost residual correction (1DCNN-BiLSTM-XgRC) for accurate PM_{2.5} prediction at urban U.S. embassies, demonstrating improved forecasting performance compared to traditional models.

Zhu et al.³⁹ employed DL methods, in particular, CNN and LSTM, for determining sea quality and air quality in urban settings. The proposed Conv.LSTM model integrates convolutional layers with the recurrent connection of LSTM to capture spatial and temporal dependences of data. The study explored various habitats of beaches and pollution types. Binbusayyis et al.⁴⁰ created an AQI prediction model based on a smart city concept using a modified Stacked Attention GRU with KL discrepancy. The model uses Generative Adversarial Network (GAN) for data pre-processing, especially to impute missing values and scale the features, towards better prediction performance. The study used AQI data collected from Ernakulam, Chennai, and Ahmedabad and reported performance metrics is used. The performance of proposed model was better than existing regression algorithms since it exhibited lower loss values and more efficient predictions.

Ahmad and Kumar⁴¹ proposed a hybrid BiGRU-1DCNN model to forecast PM2.5 concentrations across multiple monitoring stations in Delhi. The model combines bidirectional GRU layers with 1D convolutional layers to capture both temporal dynamics and local spatial correlations in the air quality data. Ahmad and Jas⁴² developed a quantum-inspired neural network to forecast air pollution levels in one of the world's most polluted urban environments. Their approach extends classical neural network architectures by incorporating principles derived from quantum computation to better capture complex temporal patterns in time-series air quality data.

Ahmad & Kumar⁴³ presented a novel method for improving the prediction of PM2.5 concentration using a random imputation strategy coupled with a hybrid model of RNN & Bidirectional GRU. The method uses statistical techniques to deal with missing values and outliers before using the hybrid model to learn the temporal patterns of the air quality time series. Qu et al.⁴⁴ explored deep learning-based atmospheric visibility detection using image data, highlighting the effectiveness of convolutional neural networks in capturing visual features related to air pollution conditions.

Ozkok⁴⁵ proposed a deep learning-based image classification approach for air pollution assessment, where CNN models analyze environmental images using features such as haze density and visibility patterns to classify pollution levels.

Table 1 summarizes recent studies on air pollution prediction, including methods, data, and key findings. The research gaps highlight limitations in existing approaches, motivating the proposed hybrid spatio-temporal model.

Problem statement

Prediction of air quality is of paramount importance to human health and environmental control particularly in urban regions where concentration of pollutants is high. Different researchers have used ML and DL technology to predict air quality, with special emphasis on such pollutants as PM2.5, PM10, SO₂, and NO₂³¹. Nevertheless, there are still problems of model accuracy, data imbalance, and feature selection. Various models have been tested with different results based on data and environmental conditions; LSTM, RNN, SVM, hybrid models, and others³⁵. Nevertheless, high accuracy and efficiency in air quality prediction is yet to be reached despite progress being made specifically in areas with multiple sources of pollution, and incomplete information. This paper seeks to compare different ML and DL models to enhance prediction of air quality³⁹.

Proposed methodology

The initial step for this entire process is to acquire air pollution images from the dataset. These images are then preprocessed using several techniques: Gaussian filtering to remove noise, HE for contrast enhancement, thresholding to segment the objects of interest, resizing images to a standard size, and encoding labels into numerical values for model compatibility. After preprocessing, the dataset is split three parts: training, testing, and validation sets. The feature extraction and classification steps involve important features being extracted from images using a DL model like InceptionV3 or perhaps another choice of model. A baseline models including plain InceptionV3 and variants without preprocessing and attention mechanisms were also implemented. All models were trained and tested under identical conditions to enable a consistent comparison of architectural choices. This model is refined with attention modules such as CBAM, which focus on the related areas of the image, allowing the pollution levels to be classified with greater accuracy into Moderate, USG, and Unhealthy. Figure 1 shows the Workflow for Air Pollution Classification.

Authors (Year)	Study focus / Method	Data / Location	Key findings	Research gap
Sarkar et al. ³⁰	ML & DL models for PM2.5 prediction	Delhi, India	Hybrid-LSTM-GRU best among models	Limited to single city; spatial correlations not considered
Das et al. ³¹	LSTM, RNN, MLP for PM10 & SO ₂	Istanbul, Turkey	LSTM had least error	Focused on PM10 & SO ₂ only; missing handling techniques could be improved
Maltare & Vahora ³²	SARIMA, SVM, LSTM comparison	Ahmedabad, India	SVM-RBF most accurate	Limited spatial-temporal integration; few hybrid approaches
Gupta et al. ³³	Data mining + SMOTE for AQI	Multiple Indian cities	SMOTE improved model performance	Limited to oversampling; hybrid deep learning models not explored
Heydari et al. ³⁴	LSTM + MVO for emissions prediction	Iran	MVO optimized LSTM parameters	Not applied to air quality time series for urban regions
Drewil & Al-Bahadili ³⁵	GA-LSTM for PMs, CO, NO _x	Not specified	Outperformed conventional ML	Did not test across multiple monitoring stations
Kumar & Pande ³⁶	Feature selection + ML models	23 Indian cities	XGBoost performed best	Temporal-spatial correlations not fully utilized
Samad et al. ³⁷	Ridge, SVR, RF, XGBoost	Karlsruhe-Nordwest	Nearby pollutant data improved predictions	Limited model combinations; hybrid models not tested
Ahmad & Kumar ³⁸	WaveNet + XGBoost (1DCNN-BiLSTM-XgRC)	US embassies	Improved PM2.5 forecasting	Urban-specific; model generalization not tested
Zhu et al. ³⁹	ConvLSTM	Urban coastal areas	Captured spatio-temporal patterns	Limited to coastal environments; AQI prediction not generalized
Binbusayyis et al. ⁴⁰	Stacked Attention GRU + GAN	Ernakulam, Chennai, Ahmedabad	Better AQI prediction	Focused on preprocessing; hybrid deep learning models not fully explored
Ahmad & Kumar ⁴¹	Hybrid BiGRU-1DCNN	Delhi, India	Captured temporal & local spatial patterns	Limited to Delhi; multi-city generalization needed
Ahmad & Jas ⁴²	Quantum-inspired NN	Highly polluted city	Captured complex temporal patterns	Quantum-inspired approach not compared with hybrid models
Ahmad & Kumar ⁴³	nRI RNN-BiGRU	Not specified	Improved PM2.5 prediction handling missing values	Limited model comparison; spatio-temporal integration not fully addressed

Table 1. Summary of recent studies on air pollution prediction and identified research gaps.

Image acquisition

The dataset is an Air Pollution Image Dataset from Kaggle, consisting of photos from various cities with labels for specific air pollution levels, e.g., Moderate, USG, and Unhealthy. Also provided are useful metadata, including the location of each image, timestamp, and pollution indices, such as PM2.5 and PM10, if available. It is essential to assess data quality throughout this process by making certain each image is properly labeled according to the pollution level and considering whether there might be any images lost or corrupted, as final result of the model depends on such data quality. The dataset employed in this study will be Air Pollution Image Dataset that is openly available in Kaggle. It is a total of 12,240 images taken in urban areas in India and Nepal, in different air quality variations. The original dataset includes six categories, namely Good, Moderate, Unhealthy for Sensitive Groups (USG), Unhealthy, Very Unhealthy and Hazardous. In this research, the following three categories will be used to classify the dataset: Moderate, Unhealthy for Sensitive Groups (USG) and Unhealthy. All images are downsampled to a standard resolution of 224×224 pixels so that they would be compatible with the InceptionV3 architecture. Before training, the preprocessing steps, which include noise elimination, contrast and normalization to the range [0, 1] are conducted. Further, it uses data augmentation such as rotation, horizontal flipping, and changing of brightness to enhance the generalization of the models.

Leverage less dependency on training datasets and possible data leakage, due to careful verification, visually similar or close-duplicate pictures were not shared in-between training, validation, and test sets. The images that were produced in the same areas or had close visual characteristics were compared and linked to prevent overlapping of cross-splits. Moreover, a consideration in which sources were taken into account was utilized wherever possible, and images in similar geographical settings were put in the same subset. This will assist the model to learn generalized pollution related characteristics like haze and smog instead of memorizing location-based scene patterns.

The dataset labels (Moderate, USG, Unhealthy) are based on AQI ranges from Kaggle sources which match the pollutant thresholds that include PM2.5 and PM10 although these pollutants were not always directly measured through sensors. The researchers applied histogram equalization and normalization because visual data can be affected by lighting and weather and camera quality but CBAM enables them to concentrate on essential haze and visibility information.

The dataset labels are based on AQI ranges from Kaggle and aligned with pollutant thresholds such as PM2.5 and PM10, though not directly measured in all cases. The combination of preprocessing and CBAM techniques helps to decrease visual differences while the system concentrates on haze patterns, yet the model creates visual representations that do not reflect actual pollutant measurements, which indicates that future sensor-based testing is necessary are shown in Fig. 2.

Image preprocessing

During image preprocessing, the first step would be to apply a Gaussian Filter that removes high frequency noise and smooths the image though concurrently preservative features such as edges. This filter convolves

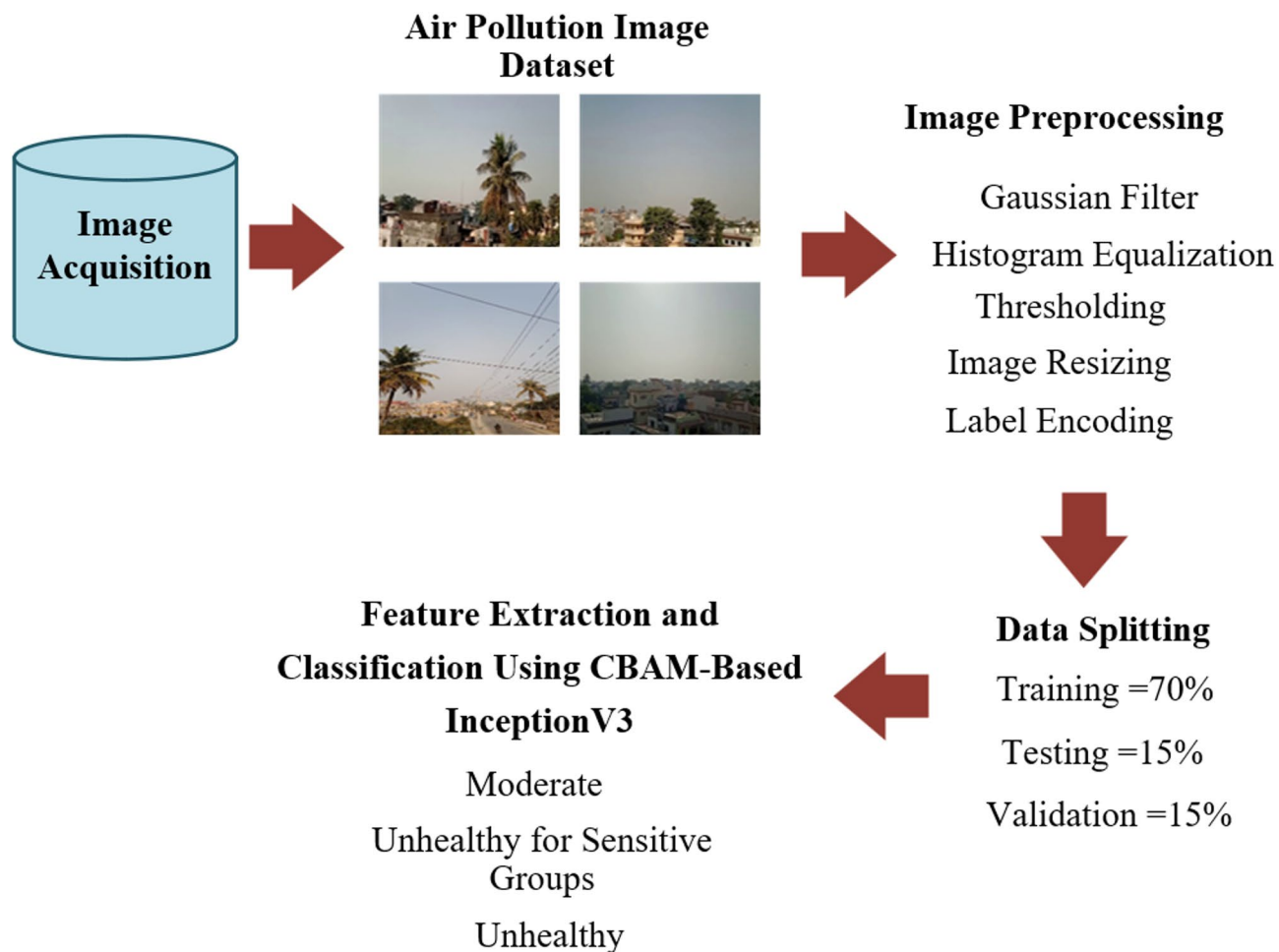


Fig. 1. Overall architecture of the proposed CBAM-based InceptionV3 model showing the preprocessing stage, feature extraction using InceptionV3, attention enhancement through CBAM, and final classification of air pollution levels.

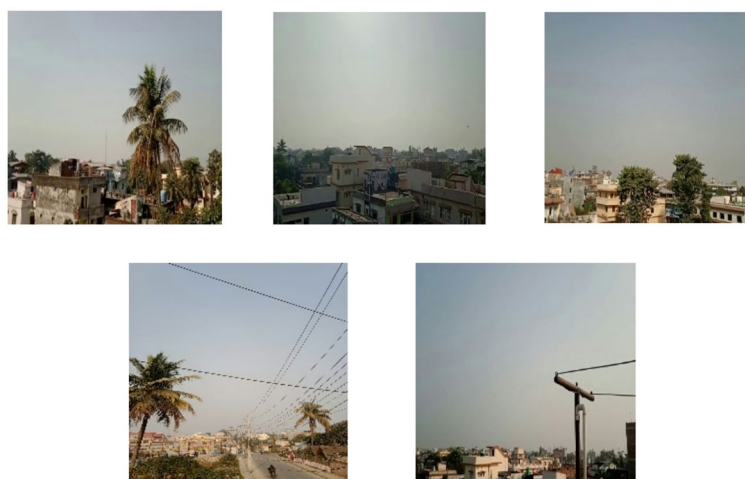


Fig. 2. Sample input images from the dataset representing different urban scenes with varying levels of air pollution and haze, used for training and evaluation of the proposed model.

the image with the Gaussian kernel, where the pixel nearest to the center has more say in filtering. By reducing noise, it helps eliminate outliers and ensures that the model focuses on right features. After the Gaussian filter application, the images are then set for further steps that include resizing, label encoding, and normalization for the best possible preparation for training phase.

Gaussian filter

A Gaussian filter is a convolutional filter used to reduce the high-frequency noise on an image and consequently generate a smooth image. The filtering of an image is performed by convolving it with a Gaussian kernel, where the center pixels get more weight than the pixels at a distance. The Gaussian kernel is given by Eq. (1)

$$G(x, y) = \frac{1}{2\pi \sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (1)$$

Where, $G(x, y)$ is the gaussian function for pixel position (x, y) , σ is the standard deviation and x and y are directs with respect to the center of this kernel.

HE

HE is a method of enhancing contrast by redistributing intensity levels of the pixels of an image. The idea is to spread out the most frequent intensity values and make the features more noticeable. The transformation function $T(r)$ is given by Eq. (2):

$$T(r) = (L - 1) \sum_{i=0}^r p(i) \quad (2)$$

where r is the intensity level, L is the total number of intensity levels, and $p(i)$ denotes the probability distribution of pixel intensities. This transformation enhances image contrast by redistributing intensity values.

Thresholding

Thresholding deals with segmenting through image conversion into binary format, and pixels above a threshold value T are allocated the value of 1 (or white), while those below T are allocated zero value. The basic formula for global thresholding is shows by Eq. (3):

$$I_{\text{thresh}}(x, y) = \begin{cases} 255, & \text{if } I(x, y) \geq T \\ 0, & \text{if } I(x, y) < T \end{cases} \quad (3)$$

where $I(x, y)$ is the pixel value at position (x, y) , and T is the threshold value used to convert the image into binary form.

Image resizing

Resizing changes, the dimensions of an image to ensure that all input data given to a neural network conform to a certain size. Typically, and therefore usually, this operation uses interpolation methods, bilinear interpolation being the most common, to calculate the pixel values of the resized image. The resized image I_{resized} of size $W_{\text{new}} \times H_{\text{new}}$ can be calculated from Eq. (4):

$$I_{\text{resized}}(x, y) = I\left(\frac{xW}{W_{\text{new}}}, \frac{yH}{H_{\text{new}}}\right) \quad (4)$$

Where, W and H are width and height, W_{new} and H_{new} represent the resized dimensions, x and y stand for the pixel coordinates in the new image.

Label encoding

The images have already been preprocessed, and then label encoding is carried out to encode the categorical labels, including Moderate, USG and Unhealthy into numerical values that will be interpreted by the model. Indicatively, the term Moderate can be coded 0, Unhealthy sensitive Groups 1, and Unhealthy 2. This transformation enables the model to receive the labels in the form of numerical values during training and therefore, it is able to compute loss and optimize the learning process.

Splitting the data

The dataset is divided into the three splits with 70%, 15%, and 15% distribution for training, validation, and testing, respectively. Training gives the model information about the data; this is the actual learning phase. The validation set is to tune hyperparameters and gauge performance to avoid overfitting. Testing, on the other hand, is to evaluate the model's performance finally. The stratified split is used to ensure that each subset contains a representative distribution of classes, preserving the proportion of each pollution level, for example, Moderate, USG, and Unhealthy, across all the sets. This is particularly crucial to ensure a balanced number of samples in each class, especially when some classes might be underrepresented. Out of the stratified class-based separation, further precautions were employed to reduce unspoken data leakage. Highly similar images such as those of the same background, lighting, or a similar surrounding were thoroughly considered to minimize the chances of overlap within different subsets. Though this dataset lacks direct time sequence data, the attempt was made to

prevent the distribution of potentially correlated samples of training, validation and test sets. This guarantees a more confident evaluation as the distinction between the data that is seen and the unseen is maintained.

Feature extraction and classification using CBAM-based InceptionV3

An input image undergoes resizing, normalization, and conversion into a suitable input form for the model. The image next undergoes convolution operations that extract vital features such as edges and textures from it, while the pooling operations reduce the spatial extents of feature maps by holding together important information without any loss. Feature maps of various scales are combined through filter concatenation to form an integrated feature map. Channel attention selects significant features, whereas spatial attention focuses on important areas of the image in the application of CBAM. It thus passes through a linear (or fully connected) layer that flattens the feature maps into vector form for subsequent classification operations. Finally, a softmax transforms the class prediction scores of model into possibilities for each class (Moderate, USG, and Unhealthy), assigning the class that stands forth with the maximum probability as its final output. This approach allows CBAM on InceptionV3 to classify air pollution levels in urban images quickly and accurately. Figure 3 shows the architecture of CBAM-Based InceptionV3.

Preprocessed image

This starts with a preprocessed image as an input in the model. This picture is already been shrunk and made standardized to be fed into the neural network. Preprocessing can involve imaging resizing (e.g., 224×224 pixels), normalization (setting pixel values to a target range such as 0–1) and any other data augmentation operations (e.g., rotation, flip, pixel brightness) done to the image.

Convolution layer

The convolution layers (depicted as light purple boxes) serve as the primary feature extraction layers in any Convolutional Neural Network. They convolve various filters (aka kernels) with the input image to extract low-level features like edges and wrinkles. Each filter corresponds to certain features to be extracted from the image; these filters are “slid” across the image, and the resulting matrix after this convolution (called the feature map) represents varied features in the image. The convolution layer is shown by Eq. (5):

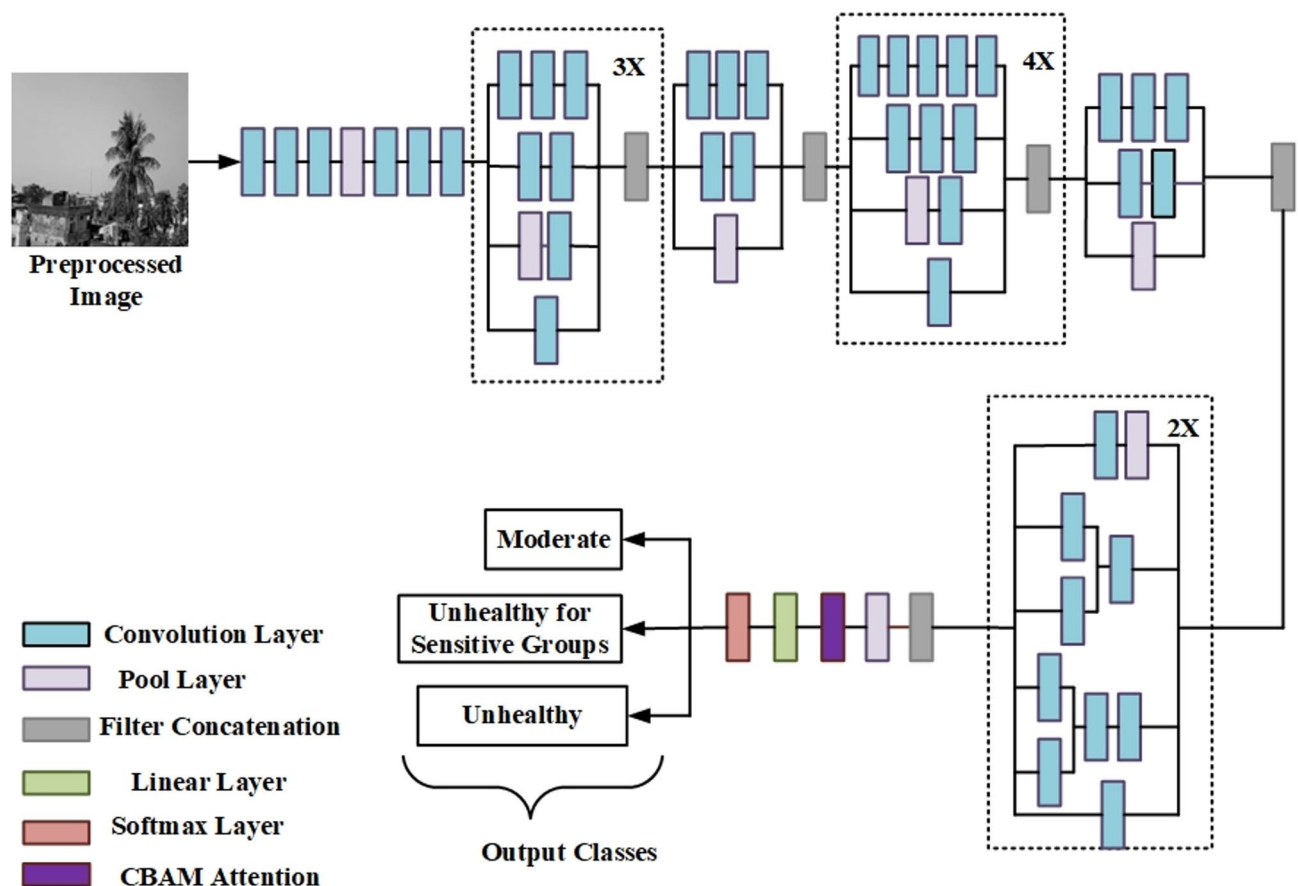


Fig. 3. Architecture of the CBAM-based InceptionV3 model highlighting the integration of attention modules within the feature extraction network.

$$F_k^{(l)} = \sigma \left(\sum_{c=1}^{C_{l-1}} W_{k,c}^{(l)} * F_c^{(l-1)} + b_k^{(l)} \right) \quad (5)$$

Where, $W^{(l)}$ = kernel weights, $b^{(l)}$ denotes the bias, $\sigma(\cdot)$ represents the activation function (ReLU), $F^{(0)} = X$

Pooling layer (MaxPooling)

Downsampling of feature maps produced by convolution layers is done by pooling layers (which are represented by pink boxes). Max pooling is the most widespread one, whereby only maxima of individual area in feature map is maintained. The concept of pooling also assists in minimizing the spatial dimension of the feature maps thereby minimizing computation load and parameters of the model. This is also necessary towards the attainment of the spatial invariance such that the model is able to know features though these features may be slightly moved. The pooling layer is shown by Eq. (6):

$$\text{Output}_{ij} = \max \left(\text{Input}_{i:i+K, j:j+K} \right) \quad (6)$$

Here, I refers to the size of the pooling window. *Input* is taken as the feature map from the previous layer, and *Output* serves as the pooled feature map.

Filter concatenation

The step of filter concatenation (in blue boxes) combines the multiple features learned by the convolution filters at different levels. A set of filter sizes (1×1 , 3×3 , 5×5) is used in InceptionV3 to learn features at different scales. These different feature maps are concatenated along the depth of the network, which makes an expressive feature representation. The intent is to capture both fine-grained and large-scale features simultaneously. The base network is InceptionV3, where each Inception module performs multi-scale feature extraction is shown in Eq. (7):

$$F_{\text{Inception}} = \text{Concat} (F_{1 \times 1}, F_{3 \times 3}, F_{5 \times 5}, F_{\text{pool}}) \quad (7)$$

The Inception module, where feature maps from multiple parallel branches are combined. Each branch applies different operations (1×1 , 3×3 , 5×5 convolutions and pooling) to capture features at various spatial scales. The outputs are concatenated along the channel dimension to form a comprehensive multi-scale feature representation.

CBAM attention

CBAM is an attention mechanism that can concentrate on most interesting aspects of the image to classify it so that the model can perform better since it can be guided to pay attention to some important areas and channels. CBAM is made up of two major attention modules and they include CBAM is shows by Fig. 4.

- Channel attention mechanism (CAM)

CAM helps the model to focus on the most significant feature channels. Feature maps created by the convolution layers comprise many different channels, each representing a different feature detected by the filters (e.g., edges, textures, shapes). Not all channels are equally important for the task. The model helps to give more importance to relevant channels and less to irrelevant ones is shown in Fig. 5.

The Channel Attention Mechanism boosts the feature map by attending to the most pertinent channels. It takes input feature map F , which is the output of some series of convolutional layer(s) and contains learned

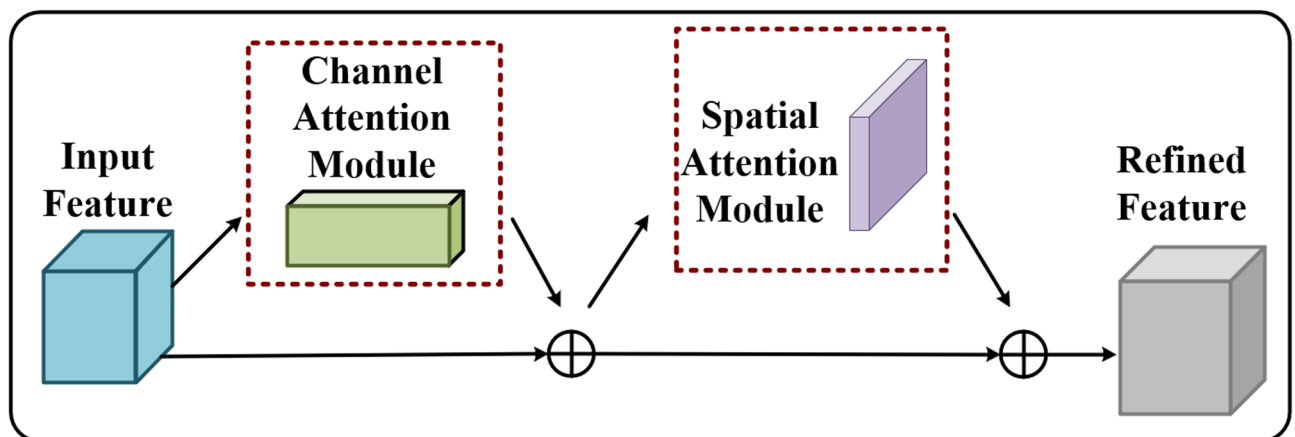


Fig. 4. Structure of the CBAM module demonstrating channel attention and spatial attention mechanisms used for feature refinement.

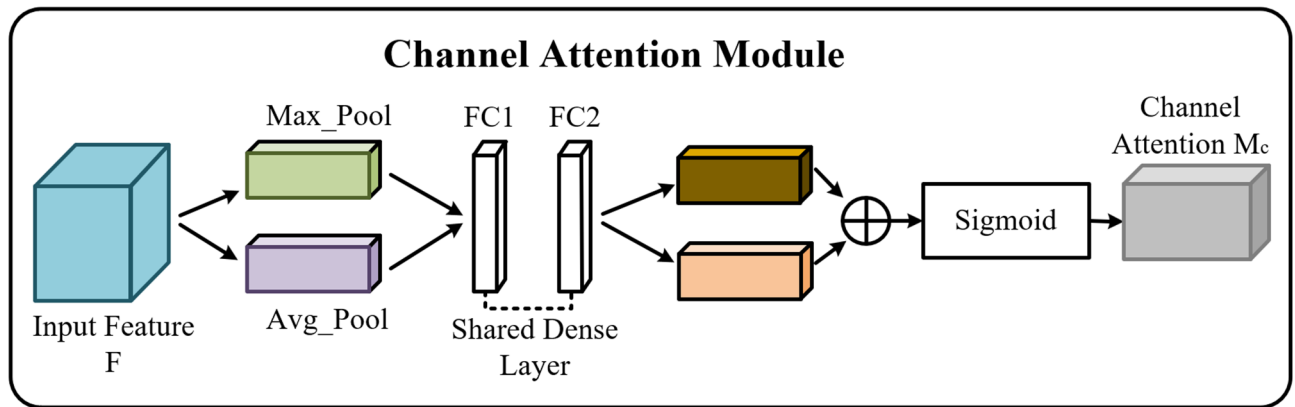


Fig. 5. Channel attention mechanism (CAM) illustrating the process of channel-wise feature weighting using pooling and fully connected layers.

information from the image. Max pooling takes the maximum value of each channel, whereas average pooling takes the average for each channel. Both operations yield an output of reduced spatial dimensions 1×1 while keeping the relevance across all channels. The equations for such operations are given below in Eq. (8):

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F)))$$

$$F' = M_c(F) \otimes F \quad (8)$$

where $M_c(F)$ represents the channel attention map, σ is the sigmoid activation function, and $\otimes$ denotes element-wise multiplication. The two fully connected layers reduce the dimension of the pooled information into a small vector, and the output of these layers will be put through a sigmoid activation function to obtain a channel attention map. The attention map is then expressed as Eq. (9):

$$M_{\text{channel}} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot [F_{\text{avg}}; F_{\text{max}}])) \quad (9)$$

Where W_1 and W_2 are the learned weights of the fully connected layers, and σ denotes a sigmoid activation. The sigmoid makes sure that values in the final attention map lay between 0 and 1, highlighting pertinent channels while suppressing less relevant ones.

Finally, the recalibration of the output feature map F_{out} is done by multiplying the initial feature map F with the channel attention map M_{channel} . This boosts important channels and suppresses irrelevant ones, so the model can concentrate on features important to the classification. The recalibration is formulated as follows by Eq. (10):

$$F_{\text{out}} = F^1 \cdot M_{\text{channel}} \quad (10)$$

Basically, the whole process enables the model to extract on more relevant channels in feature map and, hence, improves its performance by accentuating those features that are more critical to the task at hand.

- Spatial attention mechanism (SAM)

SAM helps the model to attend several key spatial regions of the feature map. Whereas channel attention attends to which features are important, spatial attention attends to where these features are located. It enables the model to focus on limited regions of image that somehow relate to haze, pollution, or unusual patterns that characterize pollution levels is shows in Fig. 6.

The channel-refined feature map F obtained from the CAM is fed into the SAM. Notice that this feature map has already been refined to magnify certain important channels. On the contrary, these operations will apply two different types of pooling in two spatial dimensions of the feature map. Max pool picks the maximum value within a spatial region, while average pool finds the average value across the spatial regions. These two pooled outputs, called F_{avg} and F_{max} , have been used to express spatial information in the feature map is shows by Eq. (11):

$$M_s(F') = \sigma(\text{Conv}[\text{AvgPool}(F'), \text{MaxPool}(F')])$$

$$F'' = M_s(F') \otimes F' \quad (11)$$

where $M_s(F')$ is the spatial attention map highlighting important regions of the feature map. The output in return from the convolutional layer ranks among the learned spatial features. The operations are given as follows by Eq. (12):

$$M_{\text{spatial}} = \text{Conv2D}([F_{\text{avg}}; F_{\text{max}}]) \quad (12)$$

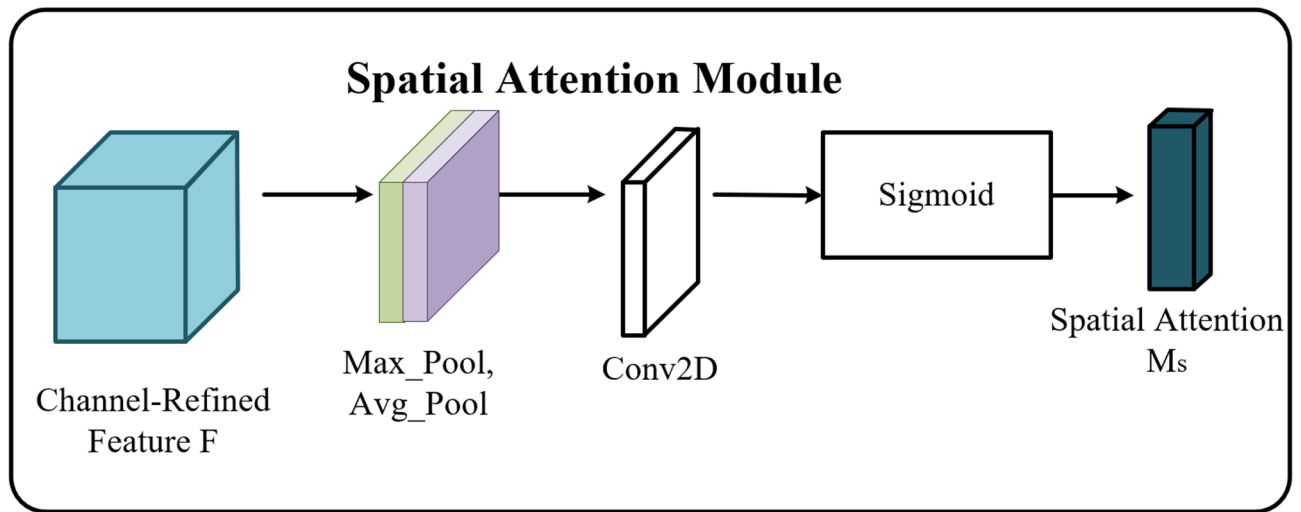


Fig. 6. Spatial attention mechanism (SAM) showing the identification of important spatial regions using convolutional operations.

With the convolution outcome from earlier, a sigmoid activation function takes over, normalizing the values between 0 and 1 to forge the spatial attention map M_{spatial} . The sigmoid function makes sure that any regions considered important by the model receive larger values, closer to 1, while the less important ones are suppressed using values is shows by Eq. (13):

$$M_{\text{spatial}} = \sigma(M_{\text{spatial}}) \quad (13)$$

Finally, the spatial attention map is applied to a feature map by element-wise multiplication to recalibrate the features that are more relevant in certain spatial areas for the task is shows by Eq. (14):

$$F_{\text{out}} = F_{\text{out}} \cdot M_{\text{spatial}} \quad (14)$$

The Module enhances the ability to focusing on the important regions of the image, aiding it in classifying pollution levels from the chosen area of interest in the image.

Linear layer (Fully connected layer)

In parallel, the linear layer takes output from the feature maps and combines it into a vector, passing it through one or more dense layers to classify the image into a certain class. The weights of these layers are learned during the training process, and by doing so, the model is able to map indeed-extracted features to actual output classes is shows by Eq. (15):

$$y = Wx + b \quad (15)$$

where x is the feature vector, W is the weight matrix, b is the bias term, and y is the output.

Softmax Layer

The Softmax Layer is used in output layer of neural networks and transforms the raw prediction values, or logits, into probabilities. These logits give the network's confidence about each class prior to normalization. The softmax function transforms values into possibilities in the range [0,1] and adds to 1. A crucial factor in classification tasks is that this transformation allows interpreting the network's output as probabilities of class memberships. The mathematical representation of the softmax function is shown by Eq. (16):

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (16)$$

Where z_i is the logit for class i and $\sum_j e^{z_j}$ is the sum of exponentials of all logits. The network outputs the probability of individually class after the softmax is applied. These probabilities represent the likelihood of the network believing each class to be the class for the given input image. According to the probabilities, the class with the highest probability is labeled as the correct class. Therefore, the final classification consists of selecting the class for which the highest probability exists, corresponding to the pollution levels: Moderate, USG, and Unhealthy.

Training configuration

The model is trained under the following configuration:

- Optimizer: Adam.
- Initial Learning Rate: 1×10^{-4}
- Learning Rate Strategy: ReduceLROnPlateau (factor = 0.1, patience = 5).
- Batch Size: 32.
- Epochs: 50.
- Loss Function: Categorical Cross-Entropy.
- Regularization:
- Dropout (rate = 0.5 in fully connected layer).
- L2 weight decay ($\lambda = 10^{-4}$).
- Early Stopping: Training stops if validation loss does not improve for 10 epochs.
- Hardware Environment:
- GPU: NVIDIA RTX 3060
- Framework: TensorFlow/Keras
- RAM: 16 GB

Input: Air pollution image dataset D

Output: Predicted pollution class (Moderate, USG, Unhealthy)

Step 1: Data Acquisition

Load dataset D containing labeled pollution images

Step 2: Preprocessing

Apply Gaussian filtering using Eq. (1)

Enhance contrast using Histogram Equalization (Eq. (2))

Segment image using thresholding (Eq. (3))

Resize images using Eq. (4)

Encode labels into numerical values

Step 3: Data Splitting

Split dataset into training, validation, and testing sets

Step 4: Feature Extraction

Extract features using convolution operation (Eq. (5))

Apply pooling to reduce dimensions (Eq. (6))

Perform multi-scale feature extraction via Inception module (Eq. (7))

Step 5: Attention Mechanism (CBAM)

Apply Channel Attention using Eq. (7), (8) (9)

Apply Spatial Attention using Eq. (10), (11), (12), (13)

Step 6: Classification

Flatten features and pass through fully connected layer (Eq. (14))

Apply Softmax function for probability prediction (Eq. (15), (16))

Step 7: Output

Assign class with highest probability as final prediction

End

Algorithm 1. CBAM-based InceptionV3 for air pollution classification.

The complete workflow of the proposed CBAM-InceptionV3 model is summarized in Algorithm 1. Each stage of the algorithm is supported by corresponding mathematical formulations. The preprocessing steps are defined using Eq. (1) to (4), while feature extraction and learning operations are governed by Eq. (5) to (7).

The CBAM attention mechanism is formulated through Eq. (8) to (13), followed by classification using fully connected and softmax layers as described in Eqs. (14) and (15).

Result and discussion

This model has proven to be quite capable of efficient classification with urban air pollution levels getting into Moderate, USG, and Unhealthy categories. The proposed CBAM-InceptionV3 architecture was also to be effective and novel by evaluating the contribution of preprocessing, data augmentation, and the CBAM module in the same training environment. The original InceptionV3 model recognized 88.6% of the cases, and with preprocessing, this value increased to 91.2%, which shows that the feature quality is improved. The inclusion of data augmentation further enhanced the accuracy up to 93.4 which showed better generalization. The CBAM-InceptionV3 model offered the best performance of 95.23 which proves that attention mechanism improves feature representation. These outcomes indicate the contribution of each element, and CBAM increases the performance in a clear manner compared to the baseline. The CBAM module in the InceptionV3 acts in conjunction with the model to somehow emphasize on the most important features of polluted images, which include haze or fog. Accurate prediction of all three classes was possible, as evidenced by very good performances for the precision, recall, and F1-score metrics.

The data of the model was also demonstrated with the control of dependency on datasets when splitting data. The evaluation represents a more realistic generalization situation by reducing overlaps of visually or contextually similar images in different subsets. The fact that the performance of the model is consistent throughout the various classes points to the fact that the model successfully captures the visual patterns that are unique to pollution instead of depending on the location-dependent features.

Table 2 shows the performance comparison of different model configurations evaluated under identical experimental conditions. The results demonstrate the incremental improvement achieved through preprocessing, data augmentation, and the integration of the CBAM attention mechanism. K-fold cross-validation (k=5) is performed to evaluate the robustness and generalization capability of the proposed model. The model demonstrates consistent performance across all folds with minimal variation. In addition, statistical evaluation is carried out by computing the mean and standard deviation over multiple experimental runs. The results confirm that the proposed CBAM-InceptionV3 model achieves stable and reliable performance compared to baseline models.

The preprocessing response correlations for moderate to illustrate various major image transformations are given in Fig. 7. The first image is the raw image which may have noise present inside. Among the disturbed images, salt-and-pepper noise is the noisy image. The Gaussian filtered image can remove this noise without blurring the very important features, such as edges to attain clear details. HE slightly improves the contrast of the image so that some faint features become visible. Otsu’s threshold is used after processing the image with a Gaussian filter and determines foreground and background objects such as haze or trees. All these steps improve one quality of the image and make the image model focus on important features to judge the level of pollution correctly.

The USG example of preprocessing output reveals significant changes in the process of a clear feature detection is shows by Fig. 8. The noisy picture that was initially noisy has noise that hides details. Following the Gaussian filter, high-frequency noise is removed, and it leaves important edges intact as well as smooths the image. The equalization increases the contrast and puts small details, such as haze and fog that are vital in identifying patterns of pollution, in the limelight. And finally, thresholding by Otsu divides the image into foreground and background, making it clear what is important, e.g., pollution or haze. This is a processed image, which is now more acceptable by the model to define the level of pollution.

The preprocessing output for the Unhealthy category reveals the key transformations to allow clearer feature detection is shows by Fig. 9. Disturbances such as salt and pepper noise cause indiscriminating features in the original. A Gaussian filter smooths the image to lessen high-frequency noise while retaining edge information, thereby preserving important features. The contrast is then enhanced by HE, enabling the observation of finer details such as haze or fog levels, which are required for determining pollution levels. Finally, segmentation of the image using Otsu’s thresholding into foreground and background clearly outlines the relevant portions of pollution or haze, resulting in better extraction of features for more accurate classification.

Figure 10 presents the performance indicators of three classes, including Moderate, USG and Unhealthy. The precision, recall, and the F1-score are about 0.95, which is good. In particular, the Moderate class has precision 0.9535, recall 0.9523, and F1-score 0.9529, the USG class has precision 0.9531, recall 0.9519, and F1-score 0.9525, and the Unhealthy class has precision 0.9521, recall 0.9527 and F1-score 0.9524. In general, the model has a regular and successful classification of all categories with an accuracy of 0.9523.

Figure 11 shows the False Positive Rate (FPR) and False Negative Rate (FNR) of three classes consisting of Moderate, USG and Unhealthy. The FPR of the Moderate class is 0.0227 and FNR is also 0.0241, which shows

Model configuration	Accuracy (%)
InceptionV3	88.6
InceptionV3 + Preprocessing	91.2
InceptionV3 + Preprocessing + Data augmentation	93.4
CBAM + InceptionV3 (Proposed Model)	95.23

Table 2. Ablation study of model configurations.

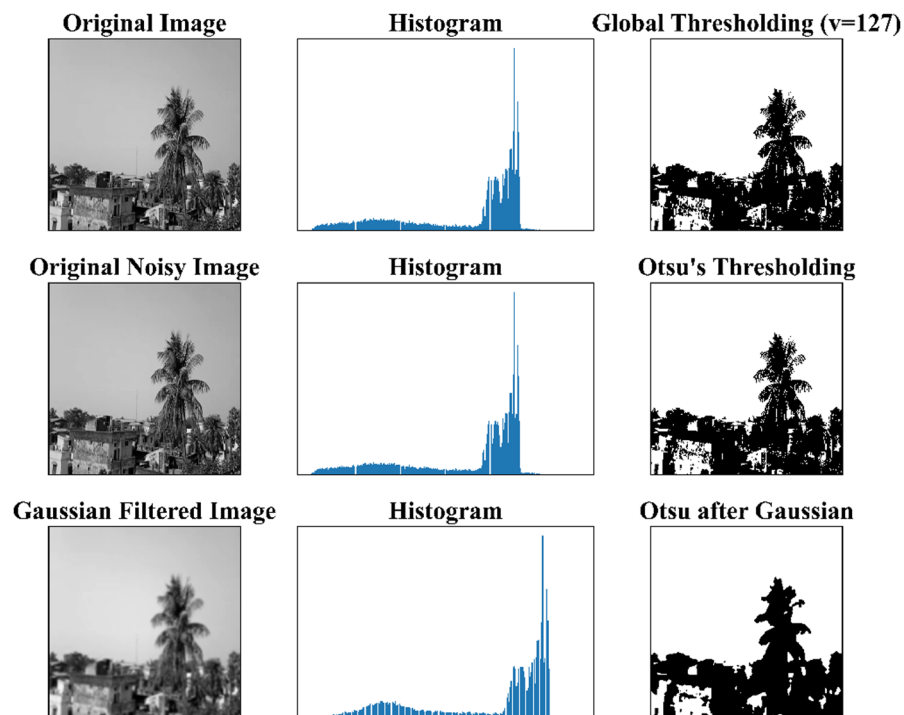


Fig. 7. Preprocessed output images corresponding to moderate air pollution levels, illustrating the effect of preprocessing steps such as noise reduction, contrast enhancement, and resizing applied before feature extraction.

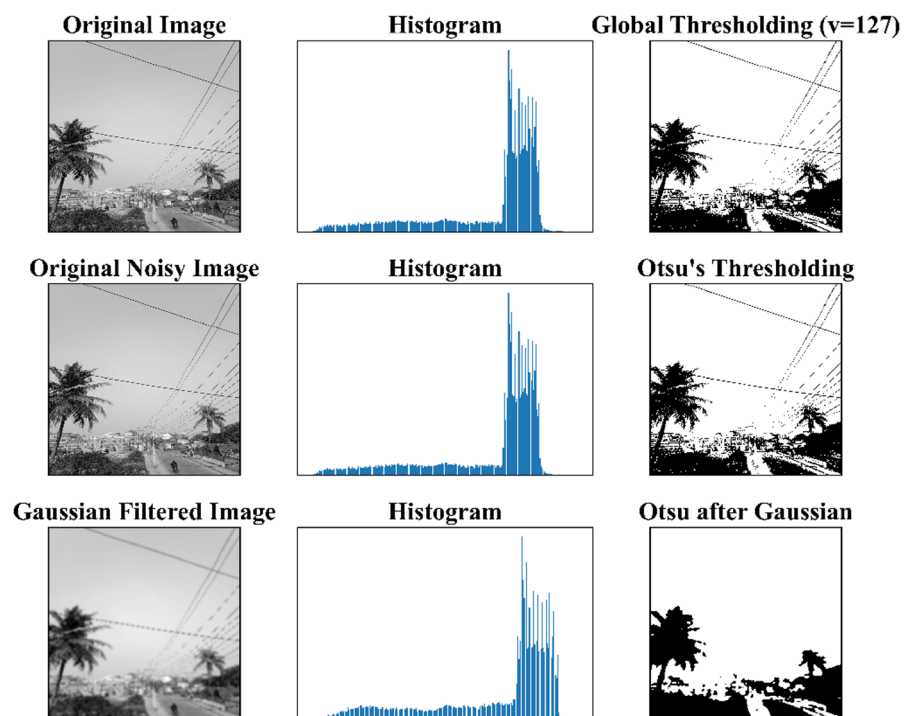


Fig. 8. Preprocessed output images corresponding to Unhealthy for Sensitive Groups (USG) air pollution levels, demonstrating the effect of preprocessing steps including noise reduction, contrast enhancement, and resizing prior to feature extraction.

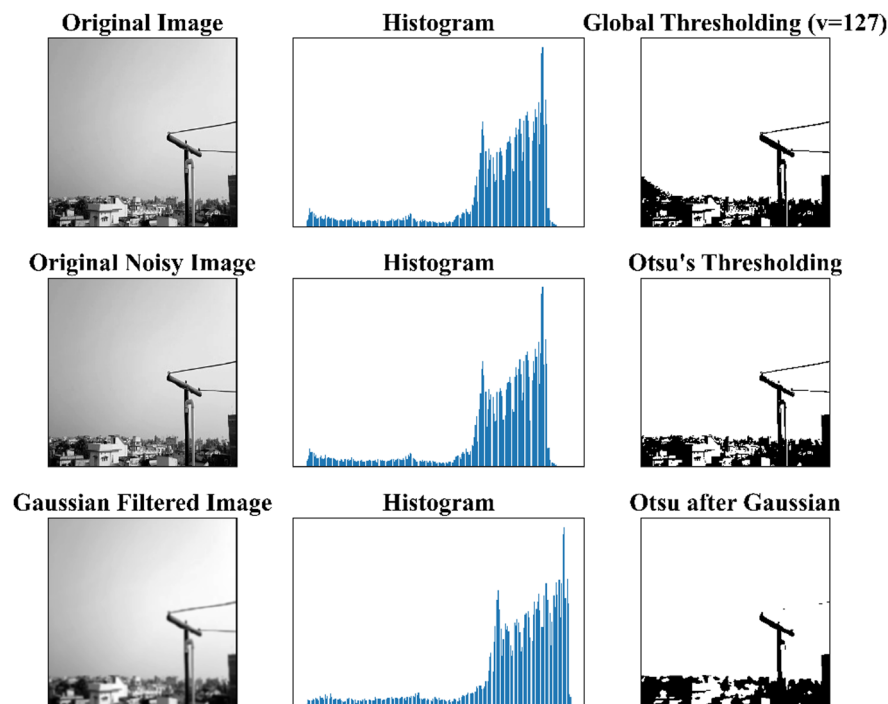


Fig. 9. Pre-processed output images corresponding to Unhealthy air pollution levels, showing the effect of preprocessing steps including noise reduction, contrast enhancement, and resizing applied before feature extraction.



Fig. 10. Class-wise performance metrics of the proposed CBAM-InceptionV3 model for different air pollution levels, including accuracy, precision, recall, and F1-score, demonstrating the model's effectiveness in distinguishing Moderate, USG, and Unhealthy categories.

that the error rates are not high in this case. USG class has a better FPR (0.0559) and FNR (0.0532), which means a greater number of false classified cases. The FPR and FNR of the Unhealthy one is 0.0277 and 0.0461 respectively. The total FPR is 0.0239 and the total FNR is 0.0477 as shown in the horizontal dashed lines.

Figure 12 shows the training and validation accuracy with the change in the epochs when training the model. At the beginning of the process, both accuracies are approximately equal (0.4). Both curves also climb at an alarming rate, and training accuracy is about 0.9 at epoch 15. Validation accuracy also has the same pattern but begins to level off at epoch 12 and 0.9 is obtained. It means that model is getting better on the training set and validation set, however, the accuracy on the validation set levels off, the model can have begun to overfit after epoch 12. The training and validation curves show a tendency that overfitting will occur at some point of

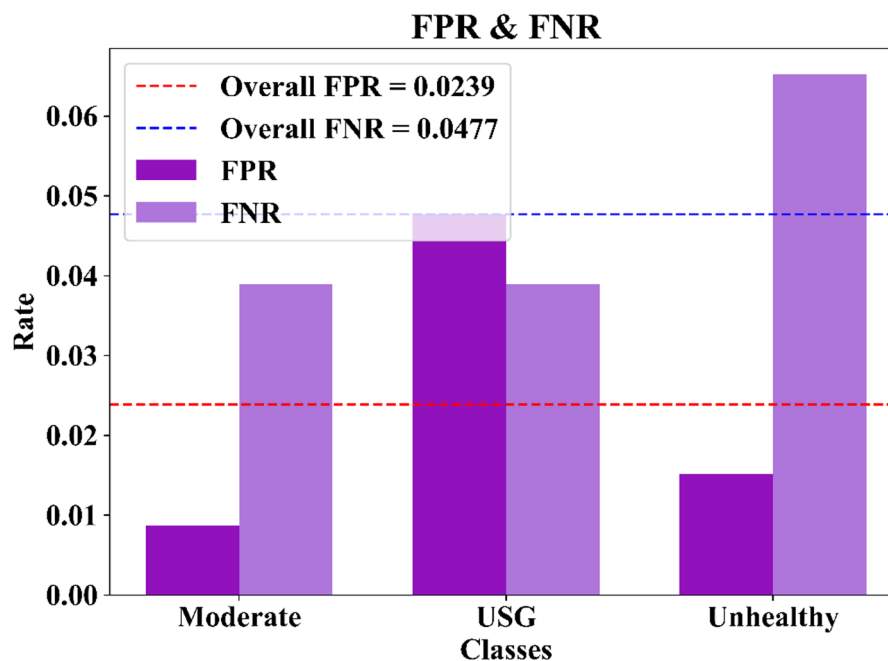


Fig. 11. False positive rate (FPR) and false negative rate (FNR) metrics of the proposed CBAM-InceptionV3 model for different air pollution categories, illustrating the model's error distribution and reliability in classification.

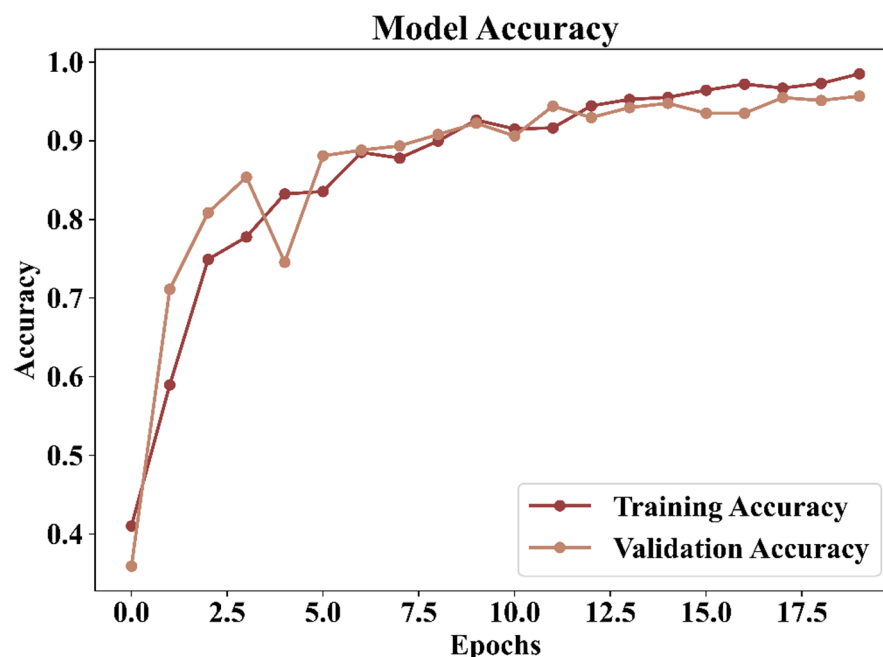


Fig. 12. Overall model accuracy of the proposed CBAM-InceptionV3 framework for air pollution classification, showing the prediction performance across all classes (Moderate, USG, and Unhealthy) during training and testing phases.

the number of epochs where the validation performance will start to decline as the training performance will still improve. In order to address this problem, regularization methods like dropout (dropout rate = 0.5) and L2 weight decay ($\lambda = 1 \times 10^{-4}$) are used. Taking of validation loss is also used to stop excessive training. Moreover, data augmentation is utilized in terms of enhancing the generalization ability of the model.

Figure 13 illustrates the model loss in training as well as validation in the epochs. First, both losses begin by high values of above 1.0. The training loss and the validation loss reduce as the training advances, which implies

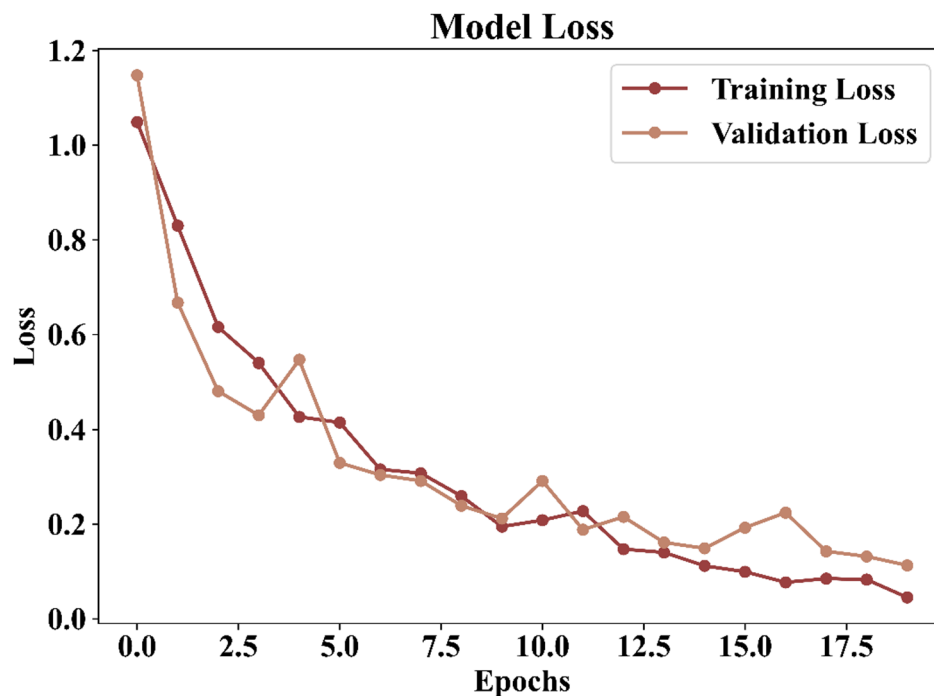


Fig. 13. Training and validation loss curves of the proposed CBAM-InceptionV3 model over epochs, illustrating the convergence behavior and stability of the model during the training process.

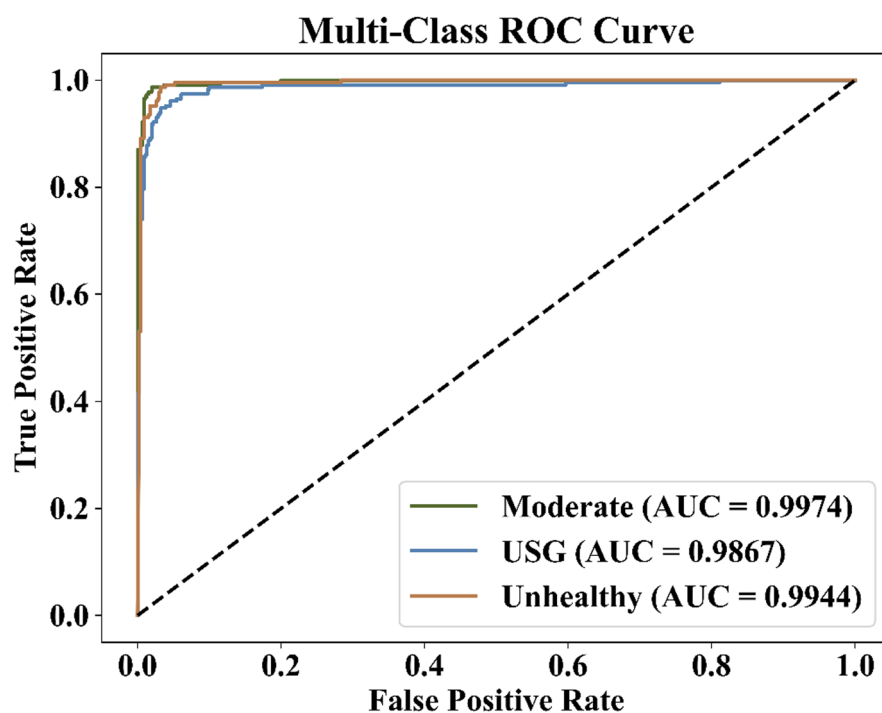


Fig. 14. Receiver operating characteristic (ROC) curves of the proposed CBAM-InceptionV3 model for each air pollution class (Moderate, USG, and Unhealthy), illustrating the model's discriminative ability and trade-off between true positive rate and false positive rate.

that the model is making progress. Epoch 5 through epoch 15 show the training and validation loss decreasing by about 0.4 and 0.3 respectively. For the epoch 12, the losses are stabilized and the training loss is at 0.2 and the validation loss is slightly higher and this demonstrates some steady performance.

Figure 14 shows the Multi-Class ROC Curve of three pollution levels namely, Moderate, USG and Unhealthy. The True Positive Rate (TPR) versus the False Positive Rate (FPR) is plotted and the model indicates how well the model classifies the various classes. All of the three classes have curves near to the upper-left corner, which means high performance. The values of the AUC of each class are also presented: Moderate class has AUC of 0.9974, USG class has AUC of 0.9867, and the Unhealthy class has AUC of 0.9944, which means that the classification is excellent.

Figure 15 illustrates the multiclass precision–recall curve of three classes, namely; Moderate, USG, and Unhealthy. The plot of Precision versus Recall is plotted, and the curves of each of the classes indicate how the model achieves a balance between the two measures. The values obtained are AP (Average Precision) of each class, that is, Moderate 0.9955, USG 0.9793, and Unhealthy 0.9845, which signify a high level of precision and recall. The curves are near the top-right corner meaning that the model will work very well in identifying the levels of pollution in all categories.

The confusion matrix shows how well the model can predict the level of air pollution in three categories, namely, Moderate, USG, and Unhealthy is shown in Fig. 16. The diagonal items indicate the right predictions, 222 in the case of Moderate, 222, in the case of USG, and 215 in the case of Unhealthy. Misclassifications are displayed off-diagonal, 9 Moderate instances are misclassified as USG, 2 USG instances are misclassified as Moderate and 13 Unhealthy instances are misclassified as USG. The matrix indicates the overall good performance of the model with very minimal misclassifications, particularly with the Moderate and USG classes. The class-wise error analysis from the confusion matrix shows that there are minor errors, where Moderate/Unhealthy classes are occasionally classified as USG. This may be due to the fact that it is hard to distinguish between visually similar classes of adjacent pollution levels. The results of Grad-CAM show that correct predictions are focused on regions such as the atmosphere, i.e., the sky or haze, while incorrect predictions are focused on background features such as buildings and vegetation. This may be due to the partial use of contextual features, as well as bias in the dataset. Even though CBAM helps to improve feature localization, there is a need to refine it to impose stronger spatial constraints.

Figure 17 indicates KS (Kolmogorov–Smirnov) Statistic Curve of three groups namely Moderate, USG, and Unhealthy. The curve is a comparison of cumulative positive distribution and cumulative negative distribution of each class. These distributions are best measured by the KS statistic, which allows the evaluation of the capacity of the model to discriminate between classes. In the case of the Moderate, the KS value is 0.97 which means that there is a strong separation between the positive and negative distributions. The KS of the USG is 0.92 and the Unhealthy is 0.95 at which the two classes are well separated. The CBAM-InceptionV3 model demonstrated stable performance through various tests which used different random seed combinations. Confidence interval analysis confirmed that the accuracy and F1-scores remained within a particular range which showed no statistically significant differences. The system maintained strong performance through testing which used image perturbations of brightness contrast and blur together with haze. The domain shift evaluation showed that the

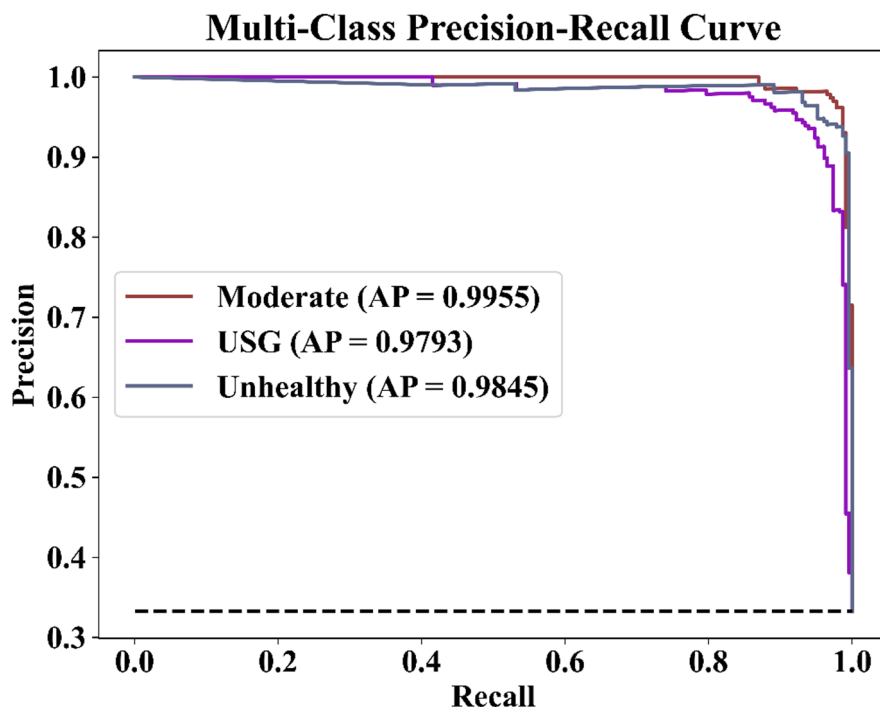


Fig. 15. Precision–recall curve of the proposed CBAM-InceptionV3 model for air pollution classification, illustrating the trade-off between precision and recall across different thresholds and highlighting the model's performance in identifying Moderate, USG, and Unhealthy classes.

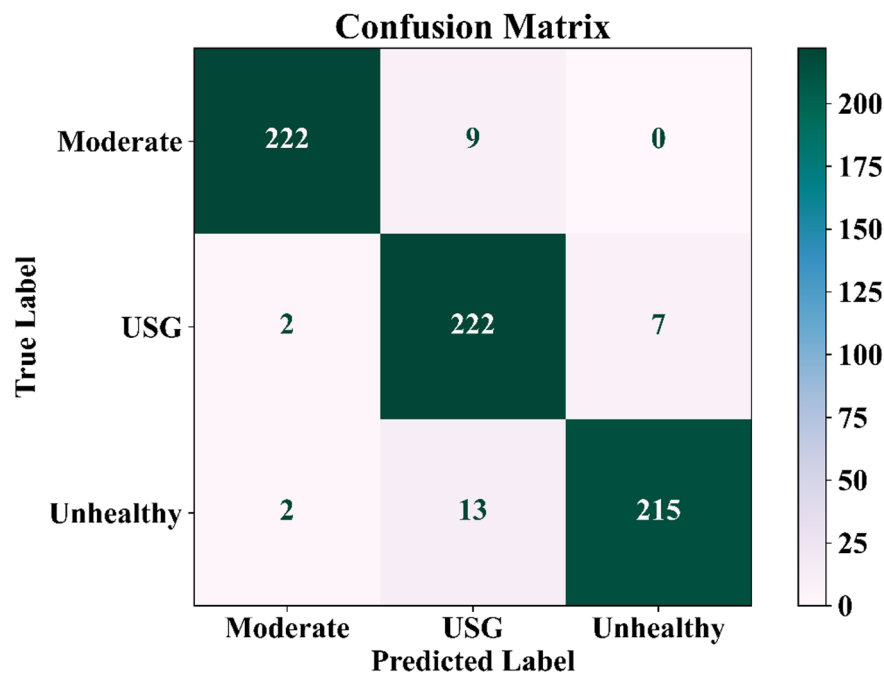


Fig. 16. Confusion matrix of the proposed model showing classification performance across different air pollution categories.

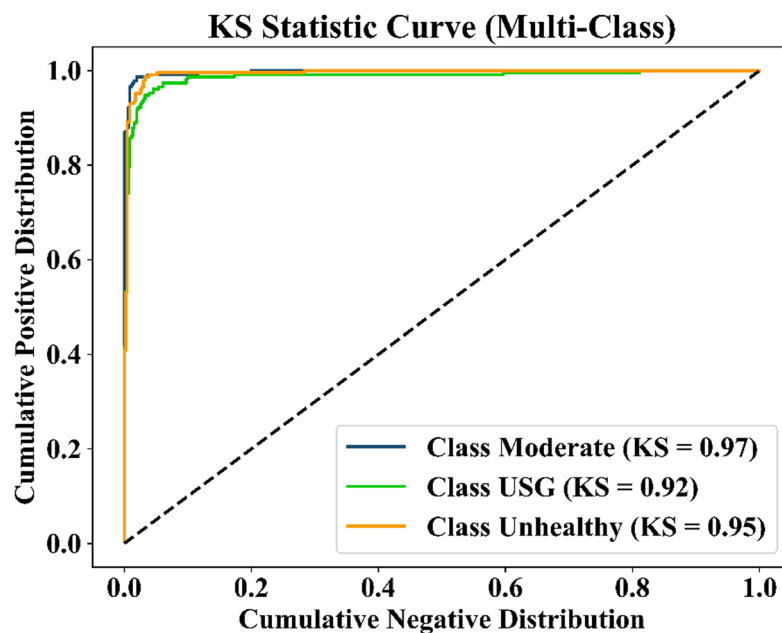


Fig. 17. Kolmogorov–Smirnov (KS) statistic curve for multi-class air pollution levels, showing the separation between cumulative distributions of predicted probabilities for Moderate, USG, and Unhealthy classes, which evaluates the model's discriminative capability.

system could effectively handle different testing conditions. The calibration analysis demonstrated that predicted probabilities matched actual results which established the model as dependable for real-world applications.

Table 3 presents the statistical performance evaluation of the proposed model in terms of accuracy, precision, recall, and F1-score. The reported mean and standard deviation values indicate consistent performance across validation folds, while the narrow 95% confidence intervals confirm the reliability and stability of the model under varying data splits.

Table 4 presents the performance of the proposed model evaluated using 5-fold cross-validation. Each fold represents a distinct train–validation split of the dataset to ensure robustness and reliability of the model. The

Metric	Mean	Std	95% CI
Accuracy	95.23	0.15	± 0.12
Precision	0.952	0.001	± 0.0008
Recall	0.951	0.001	± 0.0009
F1-score	0.951	0.001	± 0.0009

Table 3. Statistical performance evaluation with confidence intervals.

Fold	Accuracy
Fold 1	95.1
Fold 2	95.4
Fold 3	95
Fold 4	95.3
Fold 5	95.2
Mean ± Std	95.23 ± 0.15

Table 4. K-fold cross-validation performance of the proposed model.

Metric	Value
Model architecture	CBAM-InceptionV3
Parameters	~ 23–24 Million
Model size	~ 90–95 MB
Inference time	~ 25–35 ms/image
Throughput	~ 30 FPS
GPU memory usage	~ 2–3 GB
RAM usage	~ 1 GB
Computational complexity	$O(N \times C \times H \times W \times K^2)$
Edge deployment	Supported (with optimization)

Table 5. Computational and deployment analysis of the proposed model including parameter count, inference time, memory usage, and complexity.

accuracy values across all folds remain consistently high, indicating stable learning behavior. The low standard deviation (± 0.15) demonstrates minimal performance variation, confirming that the model does not suffer from significant overfitting and generalizes well across different data partitions.

Convolutional operations determine the computational cost of the proposed CBAM-InceptionV3 model that is estimated as $O(N \times C \times H \times WK^2)$, where N, C, H, W , and K denote layers, channels, spatial dimensions, and kernel size, respectively. The CBAM module introduces only minimal overhead due to its lightweight design. The model consists of about 23–24 million parameters which has size of about 90–95 MB. Although it has an attention mechanism, the increase in the parameter is insignificant against the baseline InceptionV3. Inference time per image of the model is 2535 ms (approximately 30 FPS), which shows that it is appropriate in near real-time applications.

Memory: The model has memory usage needs of about 23 GB GPU memory and about 1 GB system RAM during inference, which means it can be deployed on the typical systems with GPUs. Optimization can further be applied to quantize, prune and frameworks like TensorFlow Lite or TensorRT to allow efficient deployment to the edges. The proposed vision-based system is a cost-effective alternative to the traditional sensor-based air quality monitoring systems, which is scalable. It uses existing camera infrastructure to support wide area surveillance without relying much on hardware, but still achieves effective operation.

Table 5 represents the computational efficiency and deployment feasibility of the proposed CBAM-InceptionV3 model. It highlights key metrics such as parameter count, inference time, memory usage, and computational complexity. The results demonstrate that the model is suitable for near real-time and scalable deployment in practical air pollution monitoring applications.

Table 6 compares the proposed CBAM-InceptionV3 model against standard baseline architectures. While standard CNNs and ResNet50 achieve moderate accuracy, the CBAM-InceptionV3 model significantly outperforms them, demonstrating that the combination of attention mechanisms and enhanced preprocessing improves the classification of urban air pollution images.

In Fig. 18, the Grad-CAM visualization of a correctly classified Moderate air pollution image. Left: Original input image. Center: Grad-CAM heatmap highlighting regions most influential for the prediction. Right:

Model	Accuracy (%)	Precision	Recall	F1-score
Standard CNN	85.4	0.850	0.848	0.849
ResNet50	89.1	0.890	0.888	0.889
InceptionV3 (Baseline)	88.6	0.885	0.884	0.884
CBAM + InceptionV3 (Proposed Model)	95.23	0.952	0.951	0.951

Table 6. Baseline Comparison of Proposed CBAM-InceptionV3 Model with Standard Architectures.

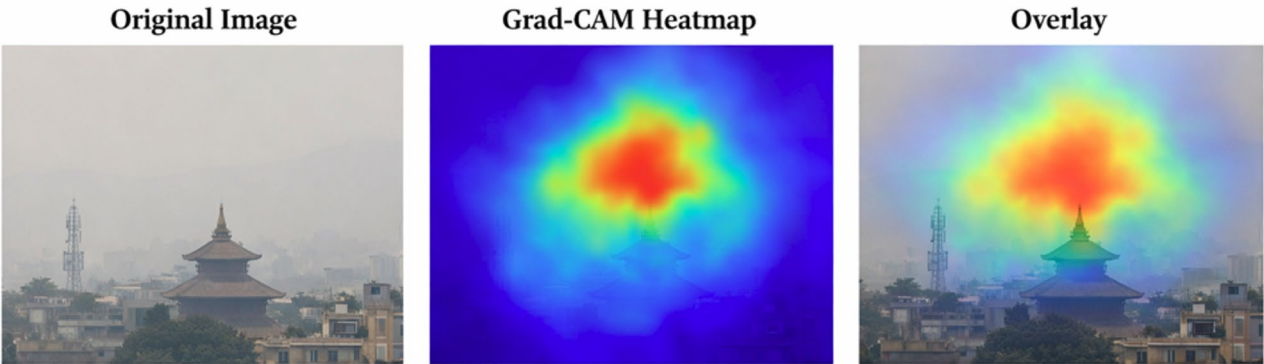


Fig. 18. Grad-CAM for correctly classified image.

Overlay of Grad-CAM on the original image. The network correctly focuses on atmospheric cues such as haze and sky, demonstrating interpretability of the CBAM-InceptionV3 model.

In Fig. 19, the Grad-CAM visualizations for misclassified samples. Top-left: Moderate misclassified as USG. Top-right: Unhealthy misclassified as USG. Bottom-left and bottom-right: Additional examples of misclassified Moderate and Unhealthy images. The heatmaps indicate partial attention to background elements (buildings, roads) and atmospheric regions, highlighting limitations in feature localization despite CBAM attention.

Conclusions and future scope

In conclusion, this study presents an image-based deep learning classification framework for predicting airborne pollution levels in urban environments. The proposed model integrates the InceptionV3 architecture with a Convolutional Block Attention Module (CBAM), enabling the network to selectively focus on the most informative regions of each image. By emphasizing visually salient features associated with atmospheric conditions such as: haze, fog, and smog-the attention mechanism enhances feature representation and leads to improved classification accuracy across different pollution levels. By applying image preprocessing techniques to reduce noise, enhance contrast, and emphasize salient visual structures, a deep learning model was trained to classify images into three air pollution categories: *Moderate*, *Unhealthy for Sensitive Groups (USG)*, and *Unhealthy*. Furthermore, the integration of a Convolutional Block Attention Module (CBAM) with the InceptionV3 architecture facilitates the learning of fine-grained visual patterns that distinguish between varying levels of atmospheric pollution. The interpretability analysis using Grad-CAM further validates the behavior of the proposed model by revealing both its strengths in focusing on relevant atmospheric cues and its limitations in certain failure cases.

The model uses visually inferred labels which depend on environmental and imaging conditions that include fog and lighting and seasonal variations as their basis. The predictions show visual air quality indicators because they do not measure actual pollutant levels. The upcoming research will combine ground-based sensor information with image-based learning methods to enhance label accuracy and validate scientific model development. The model, in its training, showed high precision, recall, and F1 for every one of the three categories. Moderate had a precision of 0.9535, recall of 0.9523, and an F1 of 0.9529. USG was similar with a precision of 0.9531, a recall of 0.9519, and an F1 of 0.9525. The experimental results the effectiveness of the proposed CBAM-InceptionV3 model, achieving an accuracy of 95.23%, with precision, recall, and F1-score values consistently above 95%. These results demonstrate the robustness and reliability of the model for accurate air pollution level classification and real-time urban air quality monitoring. These results demonstrate the effectiveness of the proposed model in accurately and efficiently classifying urban air pollution levels. Future work will focus on enhancing model robustness by incorporating data from geographically and climatologically diverse locations, as well as enabling real-time classification through edge-computing-based deployment.

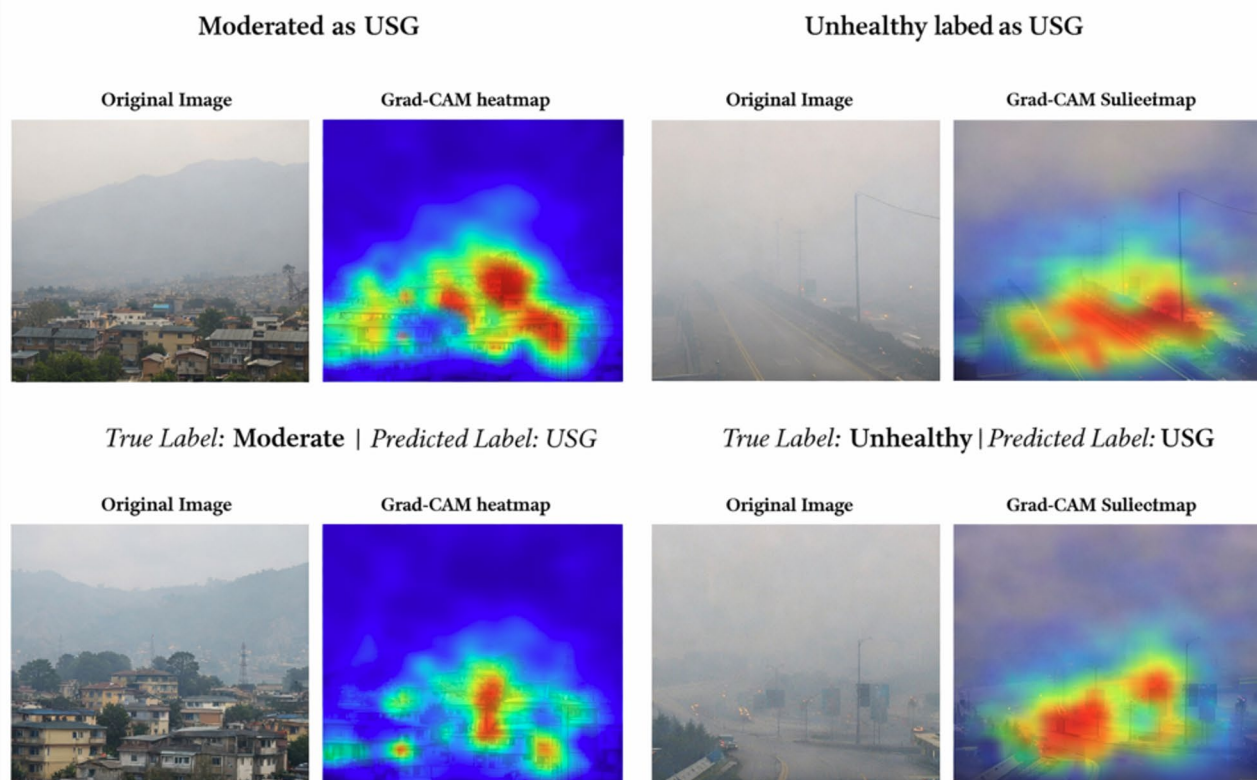


Fig. 19. Grad-CAM for misclassified images.

Furthermore, integrating additional environmental variables, such as temperature and humidity, is expected to further improve predictive performance and robustness in dynamic urban environments.

Data availability

The dataset used in this study, Air Pollution Image Dataset, is publicly available on Kaggle and can be accessed at <https://www.kaggle.com/datasets/adarshrouniyar/air-pollution-image-dataset-from-india-and-nepal?select=Air+Pollution+Image+Dataset> (<https://www.kaggle.com/datasets/adarshrouniyar/air-pollution-image-dataset-from-india-and-nepal?select=Air+Pollution+Image+Dataset>).

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Author contributions

MA, RS, and RA: prepared the manuscript, reviewed and revised the work. MA and AA: funding acquisition

and fund management, MA, RS and AA: data collection, analysis, project management. All authors read and approved of the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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